Learning Bi-directional Social Influence in Information Cascades using Graph Sequence Attention Networks

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ABSTRACT

The online information cascades have spatial and temporal characteristics. Retweets in cascades have bi-directional social influence and temporal delay, e.g., a hub node influenced by a root node will further increase the exposure of information, enhance the influence of the root node, and cause subsequent retweets after a certain time. Existing deep learning approaches mostly consider only the decay effects of social influence, ignoring the bi-directional dependency and delay effects between graphs at different moments. Therefore, a novel method is presented here, namely the **G**raph **S**equence **A**ttention **N**etworks (GSAN) , which addresses the bi-directional attention mechanism to learn temporal dynamics of social influence, as well as the cascading structure. A graph transformer block is designed to learn the complicated dependencies of spatial and temporal features in cascades. The proposed method could achieve state-of-the-art performance and large improvements in popularity prediction compared to strong baselines in real-world datasets.

CCS CONCEPTS

• **Networks** → Online social networks.

KEYWORDS

information cascades, popularity prediction, social influence

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1 INTRODUCTION

A dynamic cascade structure is formed as information or innovative ideas propagate among users through online networks [2]. The popularity prediction of online cascades has plenty of applications, such as analysis of the dissemination of rumor or extreme violent news, and evaluating the effectiveness of commercial advertisements. An example of a real-world cascade and its popularity dynamics is shown in Fig. 1. The social influence of opinion leaders has time delay and decay effects. The retweets driven by different social hub nodes may appear after a period and the influence decays after their peak points. Moreover, the nodes impacted by the hub nodes will also reveal the social roles and influence of those hub nodes in information diffusion, and affect the further dissemination process in turn. For example, node S in Fig. 2 can better perceive its social role and influence through the bi-directional attention mechanism as shown by red lines. It benefits cascade learning if a model perceives information in both forward and backward graphs.

The traditional methods developed on manually extracted features [2] and the Hawkes process suffer from non-scalability and low precision. The deep learning method is first applied by DeepCas [3] to learn an end-to-end representation of cascades. DeepHawkes [4] combines weight decay effects with recurrent neural networks to produce better performance. CasCN [1] reduces the prediction error using recurrent cascade convolutions by considering both temporal decay and cascade structure. However, the above methods only support one-direction dependency learning and ignore the social influence delay. The Transformer networks [5] have a multi-head attention mechanism, which benefits the bi-directional dependency learning of the token sequence in many NLP tasks. However, the current Transformer networks only work on the token sequence data. None of earlier works extended Transformer networks on graph sequence data, like cascades. Therefore, we design a graph transformer block that integrates graph attention with sequence attention to support the Transformer networks on graph sequence learning. Then, the graph transformer block is cooperated with the transformer encoder block to learn multi-level dependencies of cascades. The GSAN is found making significant improvements in popularity prediction compared to baselines.

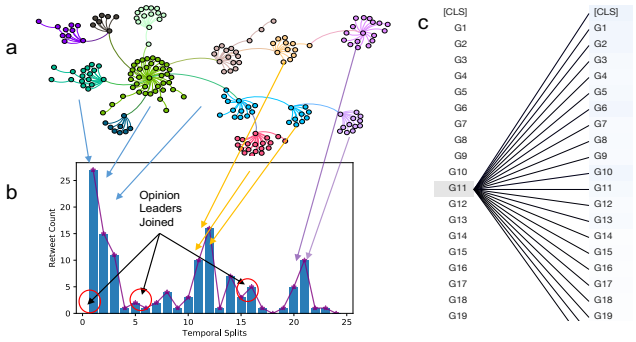

Figure 1: A cascade and its bi-directional social influence.

2 PROBLEM DEFINITION

We model a cascade as a graph sequence. The snapshot subgraph at moment $t(t = 1, 2..., T)$ forms a graph sequence, denoted as $\{G_1, G_2, ..., G_T\}$, where T is the observation time. At each moment i, the node features and graph structure are represented as $X_i \in$

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$\mathbb{R}^{N_i * F}$ and $A_i \in \mathbb{R}^{N_i * N_i}$. The target $\triangle P$ is the growth of popularity from the observation time to the predicting time. The $\triangle \hat{P}$ is the predicted value. The loss is $\sum_i (log(\triangle P_i + 1) - log(\triangle \hat{P}_i + 1))^2$.

3 THE GSAN

The whole framework of proposed GSAN is shown in Figure 2. The graph transformer block takes cascade graph structure, node encodings, and temporal encodings as inputs. For the transformer encoder block, the same structure described in [5] is used.

3.1 Graph Transformer Block

The *Graph Attention* module transforms node features according to the cascading graph structure. Then the *Attention Pool* integrates the node features to get a graph level representation at each moment. We apply both node neighbor attention and graph level attention in the block so that both local and global features are captured. Finally, a bi-directional multi-head attention is applied to learn the long-dependency relationship between the graphs. The formulation is described in the following sections.

3.1.1 Graph Attention. The node embedding features x_v are first transformed to x'_v. $x'_v = LeaklyReLu(W'x_v))$, $W' \in \mathbb{R}^{F*F'}$. Then the x'_v is aggregated with the neighboring nodes, α_{uv} is the impact weight of node u working on node v.:

$e_{uv} = attn^n(x'_u, x'_v) = x'_u W_a x'_v$.

$\alpha_{uv} = Softmax_u(e_{uv}) = exp(e_{uv}) / \sum_{k \in \mathcal{N}(v)} exp(e_{kv})$

$x_v^{update} = LeaklyReLu(\sum_{u \in \mathcal{N}(v)} (x'_u) * \alpha_{uv})$

The x_v^{update} is the output of graph attention. Note that the attention here is made different from GAT [6] to support a better parallel computation.

3.1.2 Graph Pool. All the node features are aggregated using a graph pool method, so as to produce a whole graph representation h_G. A gated mechanism is applied to calculate weights of nodes in pooling, where σ is the activation function, W_{pool} and W_g are the projection parameters.

$h_G = W_g(\sum_v \sigma(W_{pool} * x_v^{update}) \odot x_v^{update})$

3.1.3 Graph Sequence Attention . To investigate how the subgraphs are interacted and influenced. We applied the bi-directional multi-head attention [5] to learn long-dependency of the sub-graphs. We use the encodings of temporal interval to enhance the query Q, and the graph features as key K and value V, respectively, so that the spatial and temporal representations are both involved. The graph features after transformation by the self-attention mechanism are:

$attn_g^i = softmax(\frac{QK^T}{\sqrt{d_k}})V$

The multi-head attention concatenates all the self-attention features of h heads as: $Concat(attn_g^0, attn_g^1, ..., attn_g^h)W^{mh}$, W^{mh} is the projection parameter.

Since for each subgraph, the mechanism learns a set of weights influenced by other subgraphs, the method naturally learns bi-directional dependency of the subgraphs.

Finally, a sequence pooling is used to summarize the outputs of the graph transformer blocks in the way similar to graph pooling.

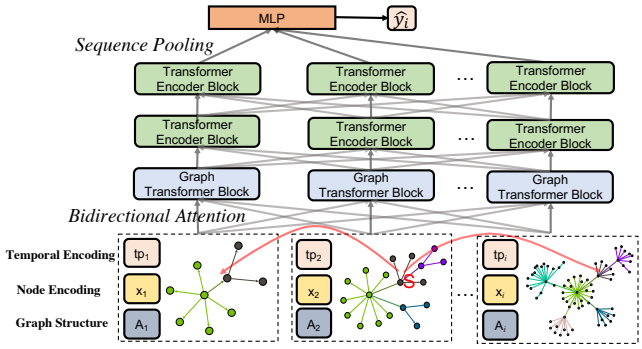

Figure 2: The framework of GSAN.

Table 1: Cascade growth prediction results on Sina Weibo.

	Microblogs			APS Citation		
T	30min	1hour	2hour	3year	5year	7year
DeepHawkes	2.77	2.32	2.23	1.74	1.51	1.36
CasCN	2.53	2.36	2.13	1.72	1.54	1.43
GAT	3.27	3.15	2.91	2.07	1.70	1.65
GSAN	**2.21**	**2.11**	**1.95**	**1.62**	**1.41**	**1.30**
Reduction	12.6%	9.1%	8.5 %	5.8%	6.7%	4.4%

3.2 Framework Advantage

Compared to previous models, the GSAN uses a bi-directional attention mechanism to capture the dependencies between graphs at each moment, producing a more reasonable and powerful social influence representation of cascades. The nodes in certain moments can perceive who impact them, how many subsequent nodes are infected by them and what kind of role they play in information diffusion, which benefits the popularity prediction of cascades.

4 EXPERIMENTS

The datasets of the Microblogs and the APS Citation in [4] are used. From the results presented in Table 1, it can seen that the GSAN has a lower error compared to those of strong baselines. The attention map in Fig. 1c shows the subgraphs in the cascade has bi-directional dependency and they has different degrees of relationship with previous and following subgraphs, corresponding to the phenomenon happens in the cascade.

5 CONCLUSIONS

We extended Transformer networks and propose graph sequence attention networks to capture bi-directional and long dependencies between the subgraphs at different moments of cascades. Experiments on real-world datasets have demonstrated that the proposed model outperforms the baselines by a relatively large margin.

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